

Retail Sales Performance Analysis

Exploratory Data Analysis & Business Insights Report

1. Executive Summary

This report presents an exploratory analysis of a retail sales dataset containing four years of transactional data. The objective of the analysis is to evaluate the company's business performance, identify sales and profit trends, understand customer and product behavior, and recommend strategies to improve profitability and business growth.

2. Business Objective

The objective of this project is to analyze the overall performance of the retail business by answering key business questions, including:

- Is the company profitable?
- Which products contribute the most to sales and profit?
- Which customer segments are most valuable?
- Which geographical regions perform the best?
- Are there seasonal sales trends?
- How does discount affect profitability?
- Which shipping methods are most commonly used?

The goal is to convert raw sales data into meaningful business insights that support strategic decision-making.

3. Dataset Overview

Attribute	Details
Dataset Name	Superstore Sales Dataset
Industry	Retail
Time Period	2023-2026
Total Records	10,194
Total Columns	21
Grain	Each row represents one product within an order
Measures	Sales, Profit, Quantity, Discount
Dimensions	Category, Region, Segment, Ship Mode, Order Date, Product, Customer

4. Data Cleaning & Preparation

Before performing analysis, the dataset was examined to ensure accuracy and consistency.

The following data quality checks were performed:

- Checked for missing values.
- Checked for duplicate records.
- Verified data types for all columns.
- Validated negative values.
- Created helper columns for time-based analysis.

Helper Columns Created:

Column	Purpose
Order Year	Yearly trend analysis
Month Number	Correct month sorting
Month Name	Monthly trend analysis
Quarter	Quarterly analysis
Delivery Days	Shipping performance analysis

5. KPI Summary

KPI	Value
Total Sales	₹2,326,534
Total Profit	₹292,297
Total Orders	5,111
Total Customers	805
Profit Margin	12.56%

The company generated over ₹2.32 million in sales with an overall profit margin of approximately 12.56%, indicating a profitable business during the analyzed period.

6. Exploratory Data Analysis

6.1 Product Analysis

Technology generated the highest sales and profit.
Furniture generated high sales but comparatively low profit.
Tables and Bookcases recorded negative profits.

6.2 Customer Analysis

Consumer contributed over 50% of total sales.
Corporate customers showed relatively stronger profitability.

6.3 Geography Analysis

West region contributed the highest sales and profit.
Central region generated lower profit despite reasonable sales.

6.4 Time Analysis

Sales increased over the four years.
Q4 was the strongest quarter.
November and December were peak months.

6.5 Shipping Analysis

Standard Class handled most orders.
Average delivery time ranged from 0 to 5 days.

6.6 Discount Analysis

Discounts above 30% generally resulted in negative average profit.